

Books to Scrape — End-to-End Data Analytics Project

Tasks completed: Web Scraping, Exploratory Data Analysis, and Data Visualization.

Dataset note: Books to Scrape is a sandbox website created for scraping practice. Its own page states that displayed prices and ratings are randomly assigned and have no real meaning.

1. Web Scraping

Python Requests and BeautifulSoup are used to navigate paginated product pages, extract book titles, prices, stock availability, ratings, and URLs, and export the result to CSV.

2. Data Cleaning & EDA

The included working sample contains 20 records. Average price is £38.05; median price is £41.38; average rating is 2.85/5. Missing values and duplicate records were checked, and prices were tested for 1.5×IQR outliers.

3. Visualization

The project includes a price-distribution histogram, rating-distribution chart, price-vs-rating scatter plot, and top-10 price chart. A Streamlit dashboard and Power BI build guide are also included.

Metric	Value
Records	20
Average price	£38.05
Median price	£41.38
Price range	£13.99 – £57.25
Average rating	2.85/5
In-stock rate	100.0%
Price-rating correlation	-0.076

Key Findings

- The sample's mean price is £38.05, while the median is £41.38.
- All 20 sample records are marked as in stock.
- The price-rating correlation is -0.076; this is an association in this sample, not causation.
- 0 price observations are flagged by the 1.5×IQR rule.
- Because the source website says prices and ratings are randomly assigned, the analysis should be presented as a demonstration of analytics workflow rather than real market intelligence.